

General Notes on Centroids by Integration

1. Definitions:
(Area Centroids)

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA}$$

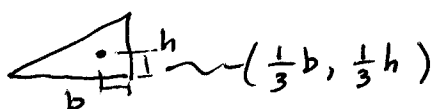
$$\bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA}$$

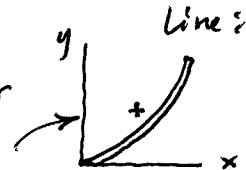
2. What is the Centroid of an Area?

Is it the "center" of an area? yes and no, depends on the area.

Center of these areas: ,  yes!

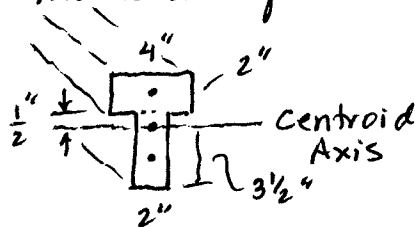
No:



Is it "on" the area or shape? Not necessarily. 

3. Centroid is a moment equation, a weighted average:

Beam Cross Section

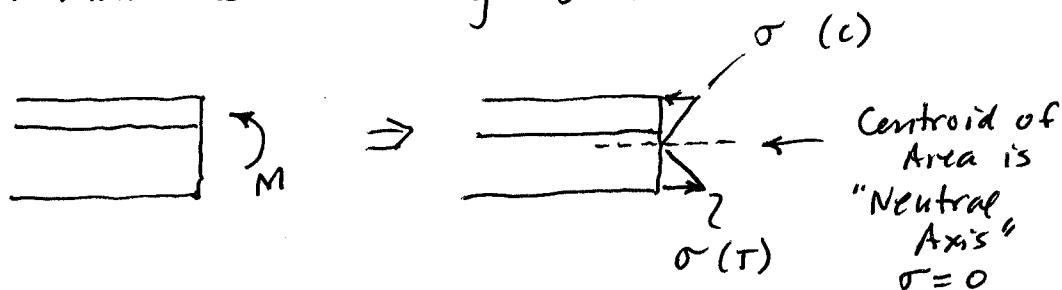


(Moment of Area) above
= (Moment of Area) below

$$8(1.5) + (.5)(2)(.25) \stackrel{?}{=} 3.5(2)(1.75)$$

$$\text{in}^3 \quad 12.25 = 12.25 \text{ in}^3$$

3. Mech of Materials: Bending Stress (σ)



Bending stresses increase linearly w/ distance y from neutral axis (at centroid).

4. Today: Centroids by Integration

Next Class: Centroids of Composite Shapes

5. Centroids by Integration

(a) Must define a dA .

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA}$$

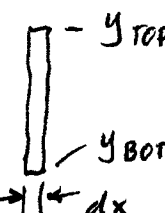
- $dA = dx dy$

- $dA = dy dx$

- $dA = \text{Horizontal Element}$

 $\frac{1}{\uparrow} dy$

Since dy thickness, must define x_L and x_R in terms of y .

- $dA = \text{Vertical Element}$

 $\rightarrow \leftarrow dx$

Since dx thickness, must define y_{top} + y_{bot} in terms of x .

(b) For double integral, limits must correspond to choice of $dA = dx dy$ or $dA = dy dx$

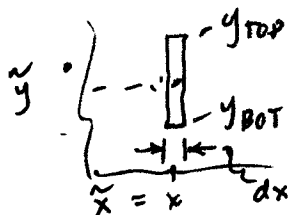
$$\int_{x_L \text{ as } f(y)}^{x_R \text{ as } f(y)} \int_{y_{bot} \text{ as } f(x)}^{y_{top} \text{ as } f(x)} dx dy$$

$$\text{or} \quad \int_{y_{bot} \text{ as } f(x)}^{y_{top} \text{ as } f(x)} \int_{x_L \text{ as } f(y)}^{x_R \text{ as } f(y)} dy dx$$

(c) Choices for $\tilde{x}$ and $\tilde{y}$:

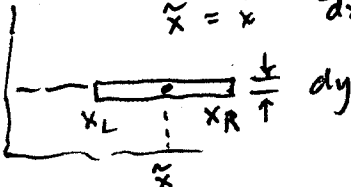
- For double integral, easy: $\tilde{x} = x$
 $\tilde{y} = y$

Regardless of $dA = dx dy$ or $dy dx$,



Vertical Element: $\tilde{x} = x$

$$\tilde{y} = \frac{1}{2}(y_{top}^* + y_{bot}^*) \quad \text{* Defined in terms of } x$$



Horiz Element: $\tilde{y} = y$

$$\tilde{x} = \frac{1}{2}(x_L^* + x_R^*) \quad \text{* Defined in terms of } y \dots$$

Centroids Equations

<u>Type</u>	<u>By Integration</u>	<u>Composite Area / line / Volume</u>
<u>Line Centroid</u>	$\bar{X} = \frac{\int_L \tilde{X} dL}{\int_L dL}$	$\frac{\sum \tilde{X}_i L_i}{L_T}$
<u>Area Centroid</u>	$\bar{X} = \frac{\int_A \tilde{X} dA}{\int_A dA}$	$\frac{\sum \tilde{X}_i A_i}{A_T}$
<u>Volume Centroid</u>	$\bar{X} = \frac{\int_V \tilde{X} dV}{\int_V dV}$	$\frac{\sum \tilde{X}_i V_i}{V_T}$

"P & G" Pappus - Guldinus Equations

$$\text{Surface Area} = A = \sum \theta \bar{r} L$$

$$\text{Volume} = V = \sum \theta \bar{r} A$$

where, θ = angle of rotation of L or A [radians]

$\bar{r}$ = distance from center of rotation to centroid of L or A

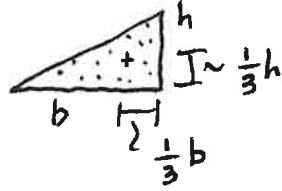
$\theta \bar{r}$ = Arc length A or L centroid moves thru

L = length of Line segment

A = Area of Area segment

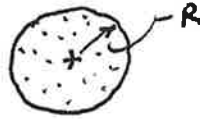
Common Areas and Centroids (And Arcs)

Triangle



$$A = \frac{1}{2}bh$$

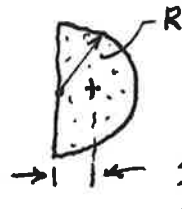
Circular Area



$$A = \pi R^2$$

$$\text{Circumference} = 2\pi R$$

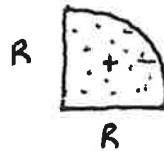
Half Circle



$$A = \frac{1}{2}\pi R^2$$

$$\frac{4R}{3\pi} = \text{Centroid}$$

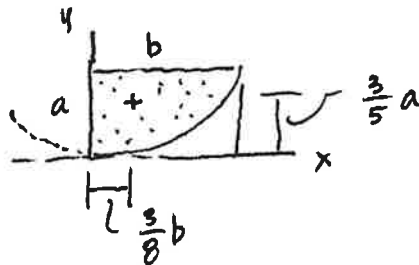
Quarter Circle



$$\frac{4R}{3\pi} = \text{Centroid}$$

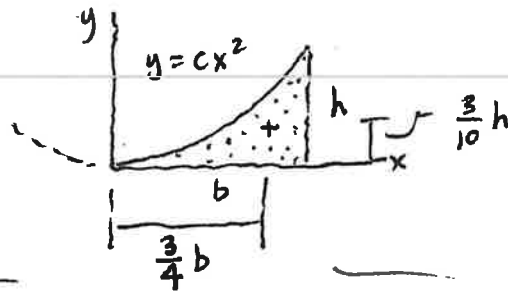
$$A = \frac{1}{4}\pi R^2$$

Parabola (n=2)



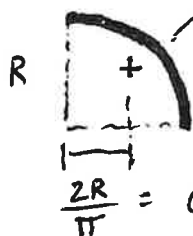
$$A = \frac{2}{3}ab$$

Spandrel (n=2)



$$A = \frac{1}{3}bh$$

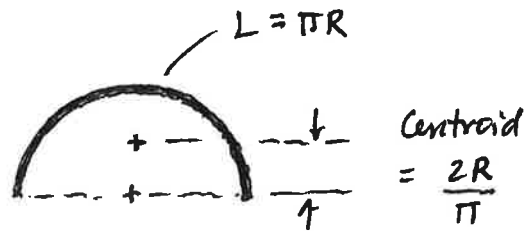
Lines (Arcs)



$$L = \frac{1}{4}2\pi R$$

$$L = \frac{\pi R}{2}$$

$$\frac{2R}{\pi} = \text{Centroid}$$



$$L = \pi R$$

$$\text{Centroid} = \frac{2R}{\pi}$$